Chapter 5 · Semester 1

CIRCULAR MOTION

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13 worked examples 2 simulations Exercises with answers

Circular motion

Circular motion

is the motion of a body **along a circular path**.

Uniform circular motion (horizontal plane)	Non-uniform circular motion (vertical plane)
a ball swung in a horizontal circle	a bucket of water swung in a vertical circle
a car or motorcycle turning a corner (flat or banked)	a roller coaster car on a loop
a conical pendulum; a merry-go-round	

Uniform circular motion

is the motion of an object in a circle with a **constant speed**: the magnitude of its velocity stays constant but the **direction of its velocity changes continually**.

Linear motion	Circular motion
displacement, s (m)	angular displacement, θ (rad)
velocity, $v = \frac{ds}{dt}$ (m s^{-1})	angular velocity, $\omega = \frac{d\theta}{dt}$ (rad s^{-1})

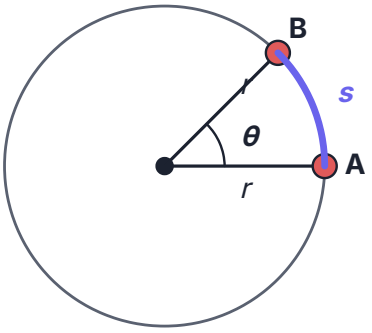
5.1 Angular displacement and angular velocity

Angular displacement, θ

is the **angle subtended at the centre by the arc length** travelled. Unit: radian (rad).

Linear distance, s

is the arc length between A and B.



$$s = r\theta$$

1 rad is the angle subtended at the centre by an arc equal in length to the radius; 2π rad = 360°.

Angular velocity, ω

is the **rate of change of angular displacement**: $\omega = \frac{d\theta}{dt}$.

- Unit: rad s^{-1}; other units: revolutions per second (rps) and revolutions per minute (rpm).
- It is a vector; its direction is perpendicular to the plane of motion (right-hand rule) and depends on whether the rotation is clockwise or anticlockwise.

Relationship between v and ω

$s = r\theta$. Divide both sides by t: $\frac{s}{t} = r\frac{\theta}{t}$

$$v = r\omega$$

The linear velocity at every point is along the tangent to the circle.

Period, T

is the time taken to make one complete revolution (unit: s).

Frequency, f

is the number of complete revolutions per second: $f = 1/T$ (unit: s^{-1} or Hz).

$$v = \frac{2\pi r}{T} = 2\pi r f \quad \omega = \frac{2\pi}{T} = 2\pi f$$

Example 5.1.1

An object moves in a circle with a uniform angular speed of 100 rpm. Calculate (a) the period, T, (b) the frequency, f.

Solution

$\omega = \frac{100 \text{ rev}}{1 \text{ min}} = \frac{100 \times 2\pi}{60} = 10.47 \text{ rad s}^{-1}$

(a) $T = \frac{2\pi}{\omega} = \frac{2\pi}{10.47} = \mathbf{0.60\ s}$

(b) $f = \frac{1}{T} = \frac{1}{0.60} = \mathbf{1.67\ Hz}$

Example 5.1.2

An object travels round a circle of radius 6 m at 30 rev min^{-1}. Calculate (a) its angular speed in rad s^{-1}, (b) its linear speed.

Solution

(a) $\omega = \frac{30 \times 2\pi}{60} = \mathbf{\pi\ rad\ s^{-1}}$ (3.14 rad s^{-1})

(b) $v = r\omega = 6\pi = \mathbf{18.85\ m\ s^{-1}}$

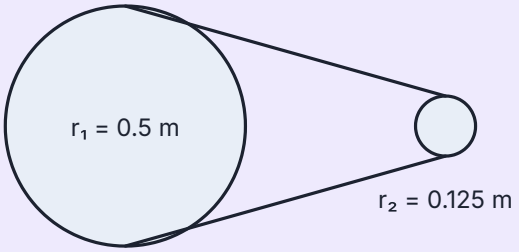
Exercises: angular velocity

1. A particle moves on a circular path of radius 0.5 m at a constant speed of 10 m s^{-1}. Calculate the time taken to complete 20 revolutions.

   Answer

   6.28 s

2. Two wheels of a machine are connected by a belt. The radius of the first wheel is 0.5 m and of the second 0.125 m. The bigger wheel turns at 3.5 Hz. What is the frequency of the smaller wheel?



Answer

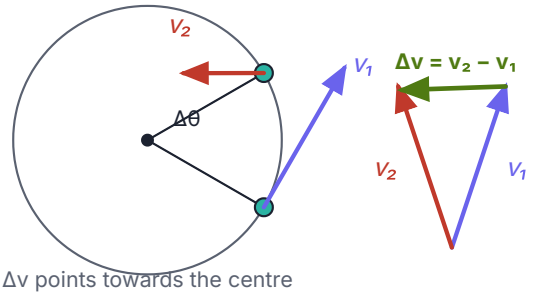
14 Hz (the belt gives both rims the same linear speed: $f_1 r_1 = f_2 r_2$)

5.2 Centripetal acceleration

An object moving in a circle of radius r at constant speed v keeps changing direction, so its velocity changes and it has an acceleration: the **centripetal acceleration**.

> **Centripetal acceleration, a_c**
>
> is the acceleration of an object moving in a circular path, **directed towards the centre of the circle**.

Direction, graphically: $a = \frac{\Delta \mathbf{v}}{\Delta t}$. As $\Delta t \to 0$, the change $\Delta \mathbf{v} = \mathbf{v}_2 - \mathbf{v}_1$ points towards the centre.



Δv points towards the centre

Additional derivation (syllabus 5.2(b)) · $a = v^2/r$

Derive the formula for centripetal acceleration.

Solution

In a short time Δt the object moves through an arc $\Delta s = v\Delta t$ and the radius turns through $\Delta\theta$.

The velocity vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ (both of magnitude v) also turn through $\Delta\theta$, so the velocity triangle and the radius triangle are similar: $\frac{\Delta v}{v} = \frac{\Delta s}{r}$

$$\Delta v = \frac{v\Delta s}{r} = \frac{v(v\Delta t)}{r} \Rightarrow a = \frac{\Delta v}{\Delta t} = \frac{v^2}{r}$$

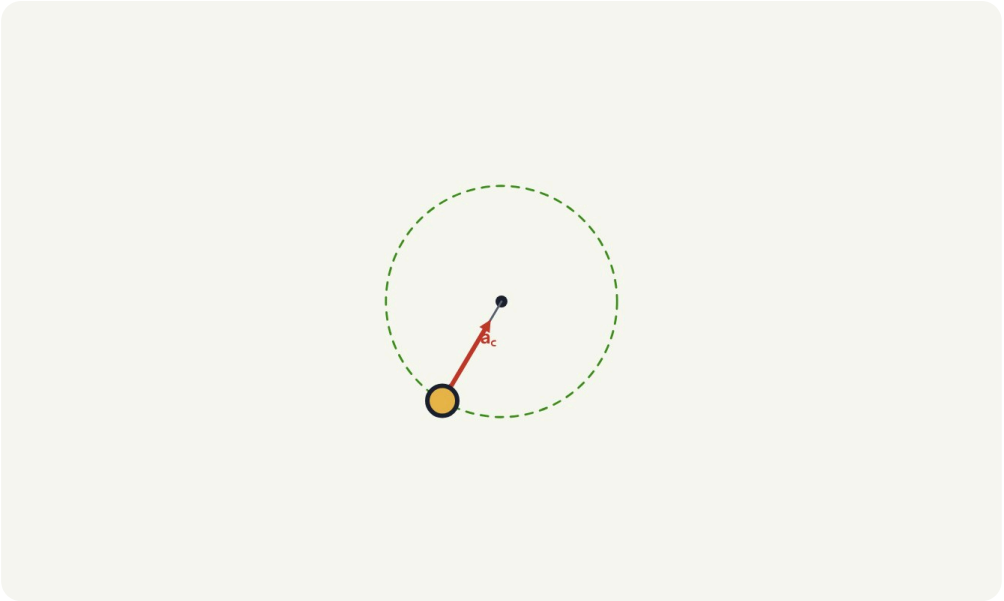
Since $v = r\omega$: $a = \frac{(r\omega)^2}{r} = r\omega^2$

$$a_c = \frac{v^2}{r} = r\omega^2 = v\omega = \frac{4\pi^2 r}{T^2}$$

Simulation

Uniform circular motion: velocity and acceleration

Change the radius and the angular velocity. The velocity arrow is always along the tangent and the acceleration arrow always points to the centre. Press "Cut the string" to see the object fly off along the tangent.



speed $v = r\omega$	period $T = 2\pi/\omega$	centripetal acceleration v^2/r	centripetal force mv^2/r
3.00 m/s	**4.19 s**	**4.50 m/s²**	**2.25 N**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 5 to use the simulation.

Example 5.2.1

Calculate the centripetal acceleration of a car travelling on a circular racetrack of radius 1000 m at 180 km h^{-1}.

Solution

$$v = \frac{180(1000)}{60(60)} = 50 \text{ m s}^{-1}$$

$$a_c = \frac{v^2}{r} = \frac{50^2}{1000} = \mathbf{2.5 \text{ m s}^{-2}}$$

5.3 Centripetal force

Centripetal force, F_c

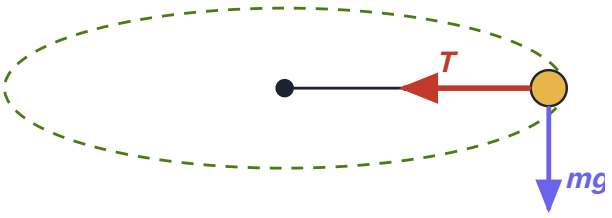
is the **net force required to keep an object of mass m moving at speed v on a circular path** of radius r.

$$F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$$

- Direction: towards the centre of the circle.
- F_c is perpendicular to **v**, so it does **no work** on the object.
- It is not a new kind of force: it is provided by another force. For the Moon orbiting the Earth, gravity provides it; for a car turning a corner, friction on the tyres provides it.

Uniform circular motion in a horizontal plane

Case 1: object whirled in a horizontal circle at steady speed. Two forces act: the weight mg, and the tension in the string. The tension is the only force in the radial direction, so it provides the centripetal force.



$$\Sigma F = ma_c \Rightarrow T = \frac{mv^2}{r}$$

Example 5.3.1

A 0.25 kg rock on a string 1.0 m long is whirled in a horizontal circle at a constant 10.0 m s^{-1}. Neglecting gravity, find the tension.

Solution

$$T = F_c = \frac{mv^2}{r} = \frac{0.25(10)^2}{1.0} = \mathbf{25\ N}$$

Case 2: a car going round a curve

(a) Flat curve. Static friction provides the centripetal force; $f_s = \mu R$.

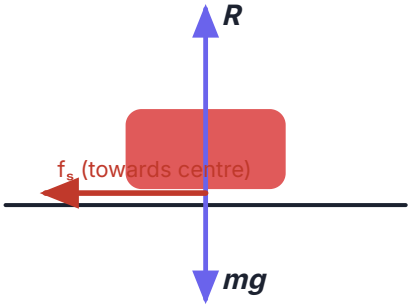
Vertical: $\Sigma F_y = 0 \Rightarrow R = mg$ (1)

Horizontal: $f_s = F_c \Rightarrow \mu R = \frac{mv^2}{r}$ (2)

(1) into (2): $\mu mg = \frac{mv^2}{r}$

$$v_{max} = \sqrt{\mu r g}$$

maximum speed without slipping



Example 5.3.2

A car travels round a flat curve of radius 50 m. μ_s between the tyres and the road is 0.75. Calculate the maximum speed without skidding.

Solution

$$f_s = \frac{mv^2}{r} \text{ and } f_s = \mu_s mg \Rightarrow v = \sqrt{\mu_s r g} = \sqrt{(0.75)(50)(9.81)} = \mathbf{19.18\ m\ s^{-1}}$$

Example 5.3.3

A 1200 kg car at 8.0 m s^{-1} goes round a flat curve of radius 9.0 m. (a) What horizontal force must the road exert on the tyres? (b) What coefficient of friction is needed for the car not to slip?

Solution

(a) $f_s = \frac{mv^2}{r} = \frac{1200(8.0)^2}{9.0} = \mathbf{8533\ N}$

(b) $\mu mg = \frac{mv^2}{r} \Rightarrow \mu = \frac{v^2}{rg} = \frac{(8.0)^2}{9.0(9.81)} = \mathbf{0.72}$

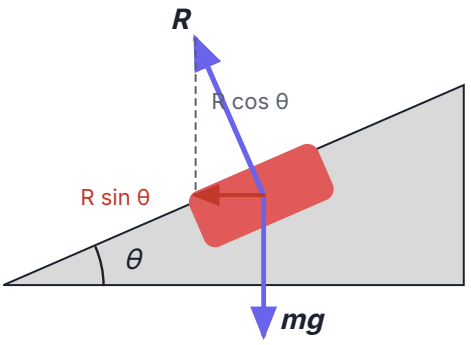
(b) Frictionless banked road. The horizontal component of the normal reaction provides the centripetal force.

x: $R \sin\theta = \frac{mv^2}{r}$ (1)

y: $R \cos\theta = mg$ (2)

(1) ÷ (2): $\tan\theta = \frac{v^2}{rg}$

$$v = \sqrt{rg\tan\theta}$$



Example 5.3.4

A curve of radius 30 m is to be banked so that a car can take it at 13 m s^{-1} without depending on friction. What must the banking angle be?

Solution

$$\tan\theta = \frac{v^2}{rg} \Rightarrow \theta = \tan^{-1}\frac{13^2}{30(9.81)} = \mathbf{30°}$$

Case 3: conical pendulum

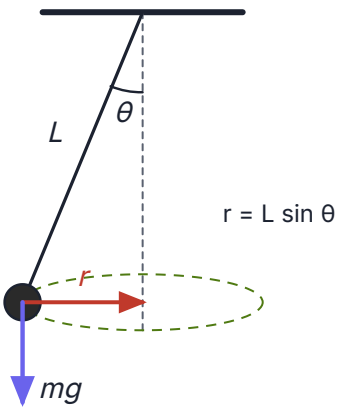
The bob moves in a horizontal circle of radius $r = L \sin\theta$. **T sin θ provides the centripetal force.**

x: $T \sin\theta = \frac{mv^2}{r}$ (1)

y: $T \cos\theta = mg$ (2)

(1) ÷ (2): $\tan\theta = \frac{v^2}{rg}$

$$v = \sqrt{rg\tan\theta} = \sqrt{Lg\sin\theta\tan\theta}$$



Example 5.3.5

A 0.15 kg ball on a 1.2 m string moves in a horizontal circle; the string makes 30° with the vertical. Find the tension and the speed of the ball.

Solution

y: $T\cos 30° = mg \Rightarrow T = \dfrac{0.15(9.81)}{\cos 30°} = \mathbf{1.70\ N}$

x: $T\sin\theta = \dfrac{mv^2}{r}$ with $r = L\sin\theta = 0.60$ m $\Rightarrow v = \sqrt{\dfrac{rT\sin\theta}{m}} = \mathbf{1.84\ m\ s^{-1}}$

Exercise: conical pendulum

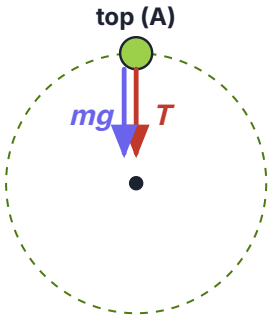
1. A pendulum bob of mass 1 kg on a string 1 m long revolves in a horizontal circle of radius 60 cm. Calculate the period of the motion and the tension in the string.

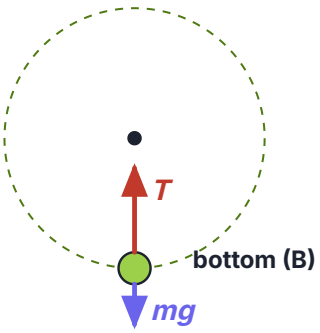
Answer

$T = 12.26$ N; period $= \dfrac{2\pi r}{v} = 1.8$ s

Circular motion in a vertical plane (non-uniform)

Case 1: a ball on a string moving in a vertical circle.





At the top (A): T and mg both point down, towards the centre.

$$T + mg = \frac{mv^2}{r} \Rightarrow T = \frac{mv^2}{r} - mg$$

T is minimum at the top.

At the bottom (B): T points up (to the centre), mg down.

$$T - mg = \frac{mv^2}{r} \Rightarrow T = \frac{mv^2}{r} + mg$$

T is maximum at the bottom.

Example 5.3.6

A 1.2 kg rock tied to a 90 cm string is whirled in a vertical circle at a constant 8 m s^{-1}. What are the tensions at the top and bottom of the circle?

Solution

m = 1.2 kg, r = 0.90 m, $\frac{mv^2}{r} = \frac{1.2(8)^2}{0.90}$ = 85.3 N

Top: T = 85.3 − (1.2)(9.81) = **73.6 N**

Bottom: T = 85.3 + (1.2)(9.81) = **97.1 N**

Minimum (critical) speed at the top

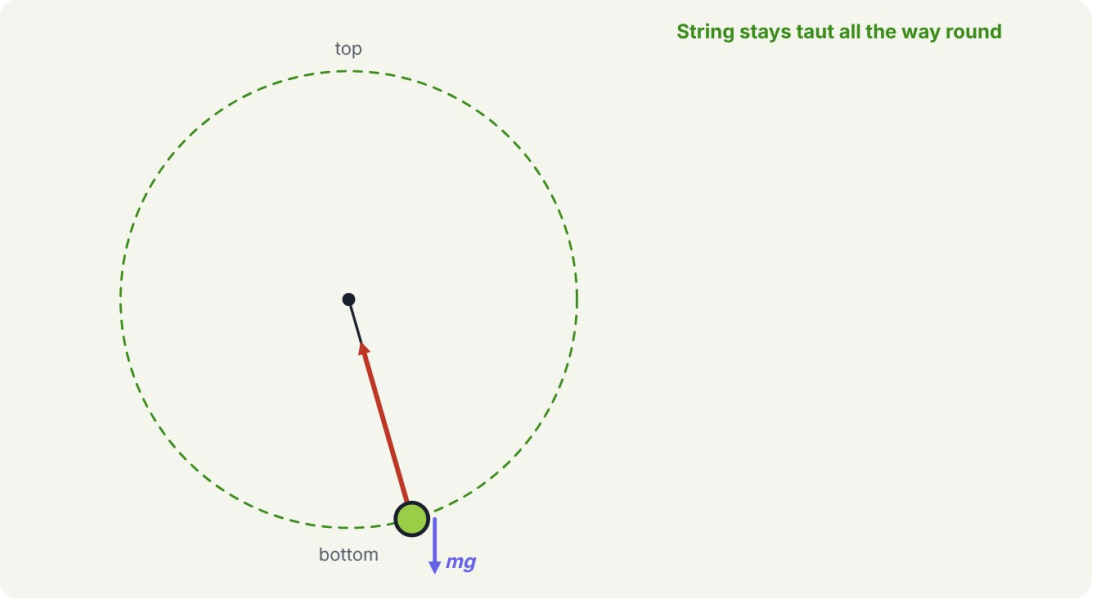
Circular motion is possible only while the string stays taut. If the string just becomes slack at the top, $T = 0$:

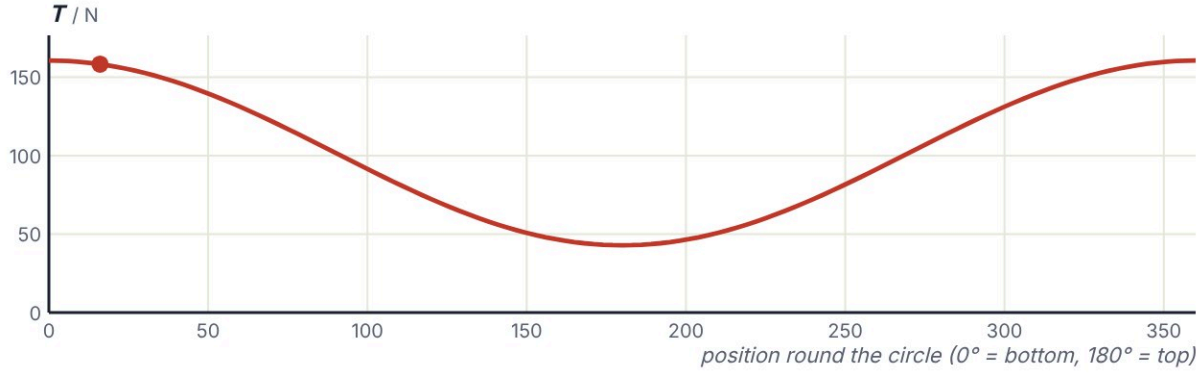
$$mg = \frac{mv^2}{r} \Rightarrow v_{\text{min}} = \sqrt{rg}$$

Simulation

Vertical circle: tension and the minimum speed

Set the speed at the top. The ball keeps its total energy, so it moves faster at the bottom. Watch the tension around the circle: below √(rg) at the top, the string goes slack.





speed now	tension now	T at top = $mv^2/r - mg$	T at bottom = $mv^2/r + mg$
7.47 m/s	**158.3 N**	**42.9 N**	**160.6 N**

minimum speed at top √(rg)
2.80 m/s

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 5 to use the simulation.

Example 5.3.7

A 2 kg ball on an 80 cm string is whirled in a vertical circle and has a speed of 5 m s^{-1} at the top. (a) What is the tension at that instant? (b) What is the minimum speed at the top needed to keep moving in the circle?

Solution

(a) $T = \frac{mv^2}{r} - mg = \frac{2(5)^2}{0.8} - 2(9.81) =$ **42.9 N**

(b) For v_{min}, $T = 0$: $v_{min} = \sqrt{rg} = \sqrt{(0.8)(9.81)} =$ **2.8 m s^{-1}**

Example 5.3.8

A bucket of water is swung in a vertical circle of radius 0.70 m. Calculate the minimum speed at the top so that the water does not spill.

Solution

The water does not spill if $T \geq 0$; at the limit $T = 0$: $v_{min} = \sqrt{rg} = \sqrt{(0.7)(9.81)} =$ **2.62 m s^{-1}**

Case 2: roller coaster on a circular track. The normal reaction R of the track replaces the tension.

Top: $R + mg = \frac{mv^2}{r} \Rightarrow R = \frac{mv^2}{r} - mg$ Bottom: $R - mg = \frac{mv^2}{r} \Rightarrow R = \frac{mv^2}{r} + mg$

A minimum speed (when $R = 0$) is needed to keep the car on the track: $v_{min} = \sqrt{rg}$.

Example 5.3.9

What minimum speed must a roller coaster have when upside down at the top of a loop of radius 8.0 m if the passengers are not to fall out?

Solution

At the top, for v_{min}, $R = 0$: $v_{min} = \sqrt{rg} = \sqrt{(8.0)(9.81)} =$ **8.9 m s^{-1}**